fish stocks were overfished (harvested beyond their biological limits) and 60 percent were maximally fished (at the maximum sustainable yield). Sustainable management is now essential if fisheries are to continue to provide jobs and meet consumer demand. The management strategies adopted depend on the nature of the problem. If fish are being taken before they are mature, it limits the stock's future ability to reproduce at a maximum level and keep numbers replenished. Placing minimum size limits on fish can help control this type of overfishing. If too many mature fish are caught, too few may be left to reproduce and replenish the present population. In this case, moratoriums and quotas are among measures that can help. Finally, ecosystem overfishing occurs when a fishery is so depleted that the ecosystem itself changes and is no longer able to support the fish stock at a sustainable level. It generally occurs when large predatory fish are overfished, allowing populations of smaller forage fish to increase and alter the entire ecosystem. This happened in the North Atlantic cod fishery: without the cod to keep them in check, the cod's three main food sources—shrimp, crab, and capelin fish—all increased in numbers.

The overfishing problem is now compounded by climate change and pollution, which are also affecting ocean ecosystems. The consequences could be dire. If global warming continues, it will cause higher ocean temperatures, sea ice will melt further, and wind and ocean current patterns will change. As a result, nutrients from the upper ocean will be transferred to the deep ocean, starving marine ecosystems and reducing photosynthesis by phytoplankton, which serve as the base food in the ocean food chain. Within three centuries—by 2300—the world's fisheries could be 20 percent less productive, and between 50 to 60 percent less productive in the North Atlantic and western Pacific. The predictions, calculated in 2018 by scientists at the University of California, Irvine, are based on extreme global warming—a 17°F (9.6°C) increase, but their models show that it is a possibility.